

A Framework for Modeling User Perceptions to Facilitate Personalized Communication

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Abstract

Technology has vastly increased the number of messages we send and receive. Despite the increasing quantity of communication, this digital communication continues to lack many of the nuances of in-person communication. In this work, I discuss my completed and planned work in modeling users' perceptions to facilitate personalized communication using untraditional modalities. I outline the general framework my work uses to develop these systems. I start by collecting participants' perceptions of arbitrary messages, and passively recording the behaviors on which the message will be based. These preferences and behaviors are then combined and communicated to the user. Their perceptions of the message are used to retrain the model to better communicate with the user. Thus far, I have used this framework to develop systems using music to promote physical activity. My planned work intends to validate the framework in a secondary paradigm, wherein a machine studies the users' behavior and adapts to their perceived needs. Overall, this research strives to develop more personalized techniques for communicating with users, striving to improve the impact and clarity of digital communication.

CCS Concepts

• **Human-centered computing** → **Interaction techniques**; **Empirical studies in ubiquitous and mobile computing**.

Keywords

Personalization, Communication, Perceptions

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1 Problem Statement

The ability to communicate drastically affects humans' daily lives. Communication shapes our interpersonal relationships [9], sense of self [35], and even society as a whole [18]. With the ubiquity of personal devices, immediate communication with others, regardless of geographical location, is widely possible for the first time in human history.



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Although technology has improved the ease of communication, the quality of digital interactions has remained rather stagnant. Digital messaging continues to lack clear affect, leading to misunderstandings between the sender and receiver [29]. Emotions can be better expressed using emojis [30], although even these expressive characters are frequently misinterpreted [3]. Multimodal digital communication, such as video calls, also lacks the nuance present during in-person interaction [10], resulting in weaker social connections than those who communicate in-person [33]. Furthermore, despite widespread use, social media often increases loneliness [25] and polarizes [34] users. Even notifications of the app, which do not involve another user, can have harmful effects, such as increasing the cognitive burden [15].

These flaws highlight a problem with modern technologies: *how can communication through and with technology be improved?* My research investigates how communication can be personalized to make a message more impactful [5, 7, 8]. I develop systems that collect and model participants' perceptions, designing personalized messages which account for how the model predicts that the user will interpret the message. Furthermore, I study these nuances through communication modalities that have been traditionally overlooked in technological communication (e.g., music). To evaluate these systems, I take a comprehensive approach, studying whether the models accurately predicted how users would interpret the message, whether the message was correctly tailored according to this prediction, and whether the users responded to the message as the system intended.

2 Related Work

2.1 Personalized Communication

Investigations of personalized communication have been conducted primarily through the lens of persuasion technology, determining how to customize messages to different aspects of the receiver's identity to maximize their receptiveness. Someone's age, agreeableness, benevolence, rational decision making, neuroticism, and extroversion have all successfully been used to persuade them [12, 36, 39]. Although personalization is often useful for persuasion, obvious personalization can "backfire" persuading someone against the intended argument. Recently, large language models have been proven to be an effective tool for constructing cheap, personalized, persuasive messages [19, 23].

Modern websites use personalized communication to deliver product and content recommendations. By the early 2000s, online retailers developed recommendation systems to suggest products to consumers [4]. Personalized marketing is so helpful that traditional marketing approaches are ineffective in comparison [31], so long as users do not feel as though their privacy has been violated [32]. Similarly, social media curates personalized experiences, providing

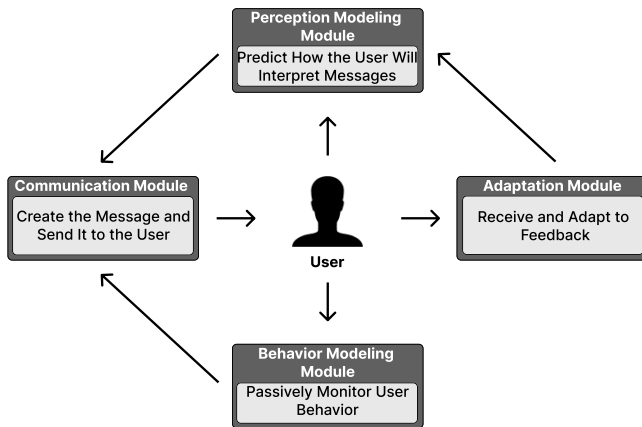


Figure 1: A high-level overview of my general framework for facilitating personalized communication by modeling user perceptions. I collect and combine users’ perceptions to tailor a customized message to the user. The models of user perceptions are adapted over time as the system learns from the user.

each user with a unique experience they will enjoy [2]. While these websites demonstrate the utility of personalized communication, they often require vast tracking and computation resources, which may be impractical for all systems.

Ubiquitous systems have also leveraged personalized communication, demonstrating that the benefits are possible without vast information about the user. MyBehavior learned user behavior through passive sensing, providing personalized recommendations to improve physical activity and diet [26–28]. Reflection Companion also tailored personalized motivational messages to each user, motivating them to be more active [17] engaged in the system [16]. Despite the success of such systems, the personalization remained relatively rudimentary, simply making references to the users’ own goals or behaviors. More complex messages could be crafted by integrating users’ perceptions into the system to provide further variety.

2.2 Modeling User Perceptions

However, research into modeling perceptions largely focuses on how users perceive systems, rather than using the perceptions as a tool for personalizing the way these systems communicate. These studies frequently investigate why users choose to trust technology. Human-robot interaction studies why a robot is perceived as trustworthy, identifying influences from a robot’s anthropomorphism [11], task completion [41], and perceived alignment with user values [1]. Similarly, AI robustness, explainability, fairness, and privacy all influence how users perceive artificial intelligence agents [14, 20]. Although understanding these influences may motivate users to engage with the system, they are unclear indicators of how systems should communicate.

Human-in-the-loop approaches have been used to help systems adapt to their users’ preferences. Systems that utilize a human-in-the-loop approach continuously collect feedback from the user, improving the performance of both cyber-physical systems [24]

and machine learning agents [40]. This continuous feedback from users has improved the performance of systems in many distinct application areas, including autonomous driving [13], collaborative robotics [21], and smart homes [37]. Such a human-in-the-loop approach could be used to learn users’ perceptions of different forms of communication, providing invaluable feedback and adjustments to such a system.

3 Methodology

My work leverages an interdisciplinary approach, combining engineering and psychology concepts to develop systems that personalize how they communicate using non-traditional modalities. My approach follows a general framework to collect users’ perceptions and tailor personalized messages accordingly. As shown in Figure 1, the framework consists of four distinct modules, each described in the following paragraphs.

Perception Modeling Module. The perception modeling module isolates factors that influence how the user will perceive the message and creates predictions of the users’ perceptions. To build these models, the module must collect baseline insight into how users will interpret various messages. To collect unbiased perceptions, the system presents the user with arbitrary example messages. Unlike the messages delivered by the communication module, these messages do not represent any actual data. Users rate each message, clearly defining how it was perceived. Once the ratings have been collected, the perception modeling module can identify relationships between the users’ interpretations and features within the message. Once these relationships are identified, the module uses them to predict how the user will respond to various messages.

Behavior Modeling Module. The behavior modeling module is the primary tool for collecting data to determine the message’s meaning. To minimize the cognitive burden on the user, the system passively observes the user’s behavior and analyzes it. For example, if the overarching system is intended to promote physical activity, this module can use a fitness tracker to monitor and record users’ steps throughout the day. In addition to monitoring the behaviors, this module should evaluate the collected data, ultimately deciding what the message delivered by the system should intend to say.

Communication Module. The communication module crafts the message, and delivers it to the user. It will collect the intended message from the behavioral modeling module and learn how to craft this specific message from the perception modeling module. Once the message is designed, the communication module is responsible for delivering it to the user. The exact implementation of the delivery is system-dependent, and could range from a UI that users can interact with to receive the message to notifications sent at optimal times.

Adaptation Module. Finally, the framework includes an adaptation module to improve the system’s understanding of users’ perceptions and refine the perception modeling module over time. This module is triggered immediately after the user interacts with the communication module’s message and asks the user for feedback about the message. Ideally, this collected feedback should be minimally obtrusive and focus on their perception of the message,

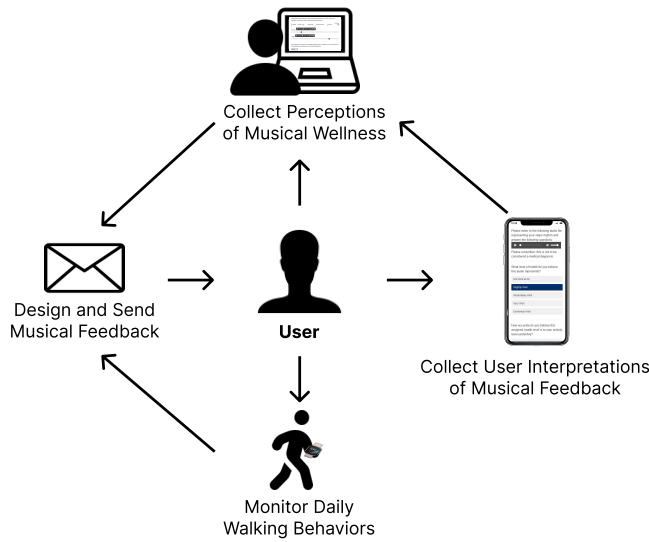


Figure 2: The system deployed in my in-the-wild evaluation. Participants completed questionnaires during onboarding that outlined their perceptions of musical wellness. Their daily activity was also recorded using Fitbits. Each day, the system constructed a melody representing their activity and delivered it to the user. They then provided their interpretation of the melody, allowing the system to compare the models’ predictions of their perceptions against the provided groundtruth. In this study, the system only compared the interpreted and intended messages and did not adjust the perception modeling module based on the feedback.

simply validating that their interpretation aligns with what the perception modeling module expects. If their interpretations do not align with the model, then the adaptation module should identify the potential confounding factors and provide this updated information to the perception modeling module. Ideally, as the feedback loop continues, the system will improve its ability to communicate with the user clearly.

4 Evaluation

I intend to evaluate my framework along two use cases: communication from a machine to a human, and from a human to a machine. My university’s Institutional Review Board approved all completed studies reported in this section.

4.1 Machine-To-Human Communication

My research into device-to-human communication explores using music to encourage physical activity [5, 7, 8]. This system paradigm is motivated by music’s unique ability to evoke emotions [38]. This system delivers physical activity feedback as music, ideally motivating participants to be more active by helping them better internalize their physical well-being.

4.1.1 Completed Work. Thus far, I have run three user studies, leading to the development of the perception modeling, behavior modeling, and communication modules.

Developing the Perception Modeling Module. I first ran a user study to develop the perception modeling module [7]. The module explored taking classical music, generated by Bach CoCoCo [22], and manipulating musical elements, such as the dynamics (volume) and tempo (speed), to communicate different messages. 55 participants were then recruited to evaluate the approach in a survey-based user study. The participants completed two short surveys, both requiring them to listen to and rate the perceived wellness of a short melody. Each survey played the same melody for every question, with musical features modified each time (e.g., one melody may be slow, while the next fast). The first survey only altered musical features one at a time, providing a baseline understanding of how participants perceived the changes. The second survey presented more complex melodies, combining the features that participants rated as sounding well and unwell in the first survey, seeking to identify whether participants’ perceptions remained stable in more complicated melodies. To ensure participants’ perceptions were not unduly influenced, participants were clearly instructed that there were “no right answers.”

Analysis of the participant’s ratings showed distinct trends between each feature and their perceptions. The most notable trend indicated that faster melodies were almost universally perceived as sounding healthier. Despite these trends, almost all participants deviated from at least one, emphasizing the importance of using personalized models. The results of the second survey confirmed that participants’ perceptions remained consistent even among more complicated melodies.

I also conducted a secondary study to inform the perception modeling module [5]. This study focused on whether chord progressions could enhance the melodies. Chords are a collection of multiple music notes played simultaneously, and are known to contribute to the emotionality of music [6]. Eight participants were recruited to create their own happy and sad-sounding chord progressions. The study intended to identify trends between the chord progressions each participant used, and the tune’s emotionality. Analysis revealed that, interestingly, while some trends existed, adherence to the trends differed based on participants’ musical background, indicating future iterations of the system must continue to offer personalized modeling.

Results from these studies informed the perception modeling module on which musical features it should analyze, and established how baseline perceptions could be collected.

Testing the Behavior Modeling and Communication Modules. Next, I developed an in-the-wild study to evaluate the behavior modeling and communication modeling modules of the system. In this study, participants wore a Fitbit and completed daily surveys that provided feedback on their physical activity. Each day, the behavior modeling module monitored the participants’ steps, and ranked their activity on a five-point Likert scale. After a week of baseline data collection with no interventions, the communication module emailed either textual or musical feedback every night at 6pm. Textual feedback stated how well the participants’ physical activity during the day was, using a 5-point Likert scale ranging from “Not Well at All” to “Very Well.” Musical feedback instead played a melody to communicate one of the five wellness levels, with the variations personalized to each user based on survey responses

collected during onboarding. Participants received each type of feedback for 31 consecutive days. Once participants listened to the feedback, the adaptation module collected information about the participants' internal state and context, and users' interpretations of the music, as they were not explicitly told what message the music was trying to communicate. 62 university students were recruited to participate in this study. The system design, consisting of all four framework modules is shown in Figure 2.

I first analyzed the number of steps participants took during each condition. The participants averaged the most steps while receiving musical feedback ($\mu = 9,260.09$), followed by textual feedback ($\mu = 8,704.38$), and then the baseline ($\mu = 8,700.17$). Although these differences were insignificant, I identified that participants were significantly more active during the second condition, regardless of the intervention order ($p = 0.047$), likely because the study was conducted during the Spring, so the weather was warmer during the second condition. Interestingly, I found that participants who received musical feedback first were not significantly more active during the second condition ($p = 0.361$). In contrast, participants who received textual feedback first were significantly more active during the music condition ($p < 0.001$). These results validated the overall system, as the participants were more active when they received musical feedback, indicating that this type of feedback may resonate with them more than traditional text. However, our results also showed that participants often interpreted the music differently than our models intended. The model's message and participant interpretations were only aligned in 25.4% of the surveys (barely better than random chance of 20%). 64.1% of responses interpreted the message to be worse than the model intended it to be. We also found that external influences, such as the participants' mood or day of the week, may influence how the music is perceived. This misalignment between users' perceptions and the perception modeling module's predictions indicate the adaptation module must learn and help the perception modeling module account for external influences.

4.1.2 Planned Work. I plan to conduct a final study for this project, specifically aimed at developing the adaptation module. This study will expand on previous results, using the perception modeling module to account for differences in perceptions across different contexts.

4.2 Human-To-Machine Communication

My work thus far has mainly focused on machine-to-human communication. I intend to continue validating my framework in a second test-case, human-to-machine communication. This study will explore how humans perceive one another, and investigate building similar internal models in a machine. With conscious awareness and investigation of potential biases, I intend to collect users' perceptions of others through ratings of actionable adjectives, such as their perceived competency. Using these insights into human cognitive processes, I plan to create models to predict the user's traits. Once these models are made, I will run a study exploring whether a device can accurately adapt its behaviors to meet the user's perceived needs in a collaborative task and whether these changes improve the team's performance.

5 Expected Contribution

Through this project, I intend to demonstrate the effectiveness of personalized communication, perception modeling, and nontraditional communication modalities. I anticipate my work will have three primary contributions. First, I introduce a framework for modeling users' perceptions to personalize communication by accounting for how they will process the message. This framework may help future systems account for these nuances, potentially improving their effectiveness. Second, my systems communicate using unconventional communication modalities. To validate these systems, my work identifies the strengths, weaknesses, and considerations that systems should account for when using these methods. These findings may help develop more accessible non-text-based systems in the future. Finally, by extracting interpretable relationships between communication nuances and user perceptions, this work will also contribute to furthering the psychological understanding of interpersonal communication.

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